Global Context-Anchored Cyclic Detector With Angle-Aware Regression for Remote Sensing Object Detection

Hui Xie, Zhanchao Huang, *Member, IEEE*, Li Wang, *Senior Member, IEEE*, and Aiguo Fei

***Abstract*—Remote sensing object detection (RSOD) is crucial for disaster monitoring, emergency response, and military reconnaissance, yet it remains challenging due to extreme scale variations and arbitrary orientations of objects. Despite recent advancements, existing methods are hindered by two critical challenges. At the feature representation level, the local progressive interaction paradigm in current feature pyramids causes semantic attenuation during long-range cross-scale transmission. At the regression supervision level, while Gaussian-based methods successfully bypass angular periodicity issues, their isotropic nature causes angular supervision to degrade when bounding boxes approach square shapes. To address these challenges, we propose the global context-anchored cyclic detector with angle-aware regression (GCAD). In particular, we introduce a global context-anchored cyclic feature pyramid network (GCAC-FPN) that employs an evolving unified context embedding spanning all scales. Through a recurrent local-to-global (L2G) and global-to-local (G2L) bidirectional interaction mechanism, this context embedding and multiscale local features continuously coevolve, enabling direct cross-scale information flow and feature refinement. Furthermore, we design an angle-aware regression (AAR) strategy that introduces explicit angular supervision into the regression loss, thereby maintaining orientation sensitivity when the Gaussian representations of square objects degenerate into isotropic distributions. Extensive experiments on two public benchmarks demonstrate that GCAD achieves state-of-the-art performance, reaching 80.72% mean average precision (mAP) on DOTA-v1.0 and 71.84% on DIOR-R.**

***Index Terms*—Angle-aware regression (AAR), global context embedding, multiscale feature fusion, remote sensing object detection (RSOD).**

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I. Introduction

REMOTE sensing object detection (RSOD) is widely applied in various fields, such as disaster monitoring, emergency response, military reconnaissance, and urban planning [1], [2], [3]. Unlike natural images, remote sensing images are typically captured from overhead perspectives via high-altitude uncrewed aerial vehicles (UAVs) or satellite platforms. This results in extensive coverage areas, which subsequently lead to complex backgrounds and considerable diversity in object orientations and scales [4], [5], as shown in Fig. 1(a). These challenges impose stringent demands on both the feature representation and regression supervision mechanisms of deep learning detectors.

This considerable scale diversity severely complicates feature extraction. Single-scale features cannot simultaneously encode fine-grained local details and broad semantic context, as progressive downsampling erodes the critical textures of small objects, while limited receptive fields disrupt the global structures of large ones. This limitation naturally motivates the adoption of explicit multiscale feature fusion strategies.

To cope with the considerable scale diversity inherent in remote sensing imagery, feature pyramid networks (FPNs) [6] and their derivative architectures have become the cornerstone of multiscale feature fusion. Researchers have endeavored to optimize FPN from various perspectives, including path augmentation, bidirectional weighted fusion, neural architecture search, and adaptive spatial fusion [7], [8], [9], [10]. Despite yielding notable improvements in bridging the semantic gap between shallow and deep feature layers, these methods remain essentially constrained by a local progressive interaction paradigm. Under this paradigm, the interaction of cross-scale information relies heavily on linear mappings or limited multipath stacking between adjacent levels, resulting in a relay-like transmission characteristic [11], [12]. As illustrated in Fig. 1(b), this relay-style transmission leads to severe semantic attenuation and feature misalignment across nonadjacent layers, which is particularly detrimental when handling the extreme scale diversity inherent in remote sensing imagery. Furthermore, even with advanced topological improvements, these methods lack an evolving unified context embedding capable of spanning all scales to synchronize feature evolution across different levels. Without such a mechanism, features at each level evolve in relative isolation, failing to achieve dynamic alignment within a global context. This lack of

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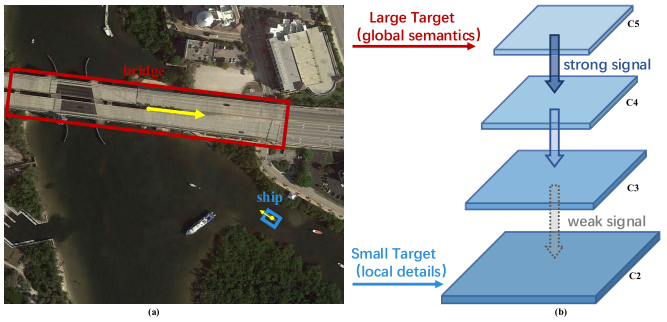


Fig. 1. Illustration of the primary challenges in RSOD. (a) Extreme scale diversity and arbitrary orientations inherent in remote sensing imagery. (b) Semantic attenuation during long-range feature interaction, where conventional relay-style top-down transmission causes severe signal decay from deep $\mathbf{C}_5$ to shallow $\mathbf{C}_2$ layers.

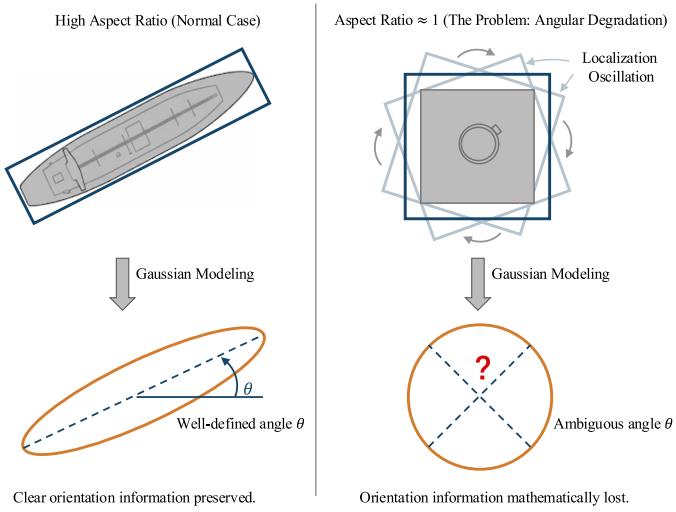


Fig. 2. Illustration of the angular degradation issue in Gaussian-based modeling. For objects with an aspect ratio of 1, the Gaussian representation degenerates into an isotropic distribution, leading to the loss of orientation information and subsequent localization oscillation.

global coherence ultimately hinders further improvement in detection performance, particularly in complex remote sensing scenarios.

Remote sensing imagery also presents a distinct challenge in object orientation. Unlike ground-level photography, aerial and satellite sensors capture scenes from arbitrary viewing angles, resulting in objects appearing at arbitrary orientations within the image plane (PL). This makes accurate orientation estimation an indispensable component of RSOD, which in turn places stringent demands on the regression supervision mechanism. In particular, remote sensing scenes are densely populated with square objects such as storage tanks (STOs), roundabouts (RAs), and baseball fields (BFs), whose geometric properties pose unique difficulties for orientation-aware regression.

Throughout the evolution of regression frameworks, two dominant paradigms have emerged. Early mainstream methodologies, such as Smooth L1 loss [13], adopt a decoupled regression paradigm that performs direct numerical regression on bounding box parameters and orientation angles. Although this approach theoretically preserves angular information, it is exceptionally sensitive to pixel-level errors and fails to resolve the boundary discontinuity problem inherent in rotation detection [14], [15], leading to suboptimal localization, particularly for objects at extreme scales. To pursue scale invariance and eliminate boundary discontinuities, loss functions based on Gaussian modeling, such as GWD [16] and KLD [17], have gained prominence. By transforming oriented bounding boxes (OBBs) into 2-D Gaussian distributions, these methods leverage divergence measures to approximate intersection over union (IoU), thereby achieving elegant boundary smoothing characteristics. However, this distribution-based paradigm introduces a latent defect when applied to remote sensing scenes. As illustrated in Fig. 2, for elongated objects, the orientation θ is clearly preserved in the covariance matrix. Yet for square objects widely present in remote sensing imagery, such as STOs and square-roofed buildings, the Gaussian representation degenerates into an isotropic distribution. In this state, the regression loss becomes insensitive to angular variations, and the resulting loss of supervision signals directly constrains localization precision for such objects.

Therefore, effectively representing multiscale features within a unified framework while maintaining robust angular supervision has become a critical bottleneck in RSOD. To address these challenges, we propose the global context-anchored cyclic detector with angle-aware regression (GCAD), which simultaneously redesigns the multiscale feature interaction paradigm and reformulates the regression supervision mechanism. The former introduces an evolving unified context embedding to bridge (BR) cross-scale information flow, while the latter explicitly integrates angular constraints into distribution-based metric learning to ensure robust orientation supervision across arbitrary object geometries. The contributions of this work are summarized as follows.

1) We propose the global context-anchored cyclic FPN (GCAC-FPN) to overcome the semantic attenuation inherent in conventional relay-style transmission. By introducing an evolving unified context embedding, we bypass layer-by-layer progressive limits and establish direct connections across all pyramid levels. Through a cyclic local-to-global (L2G) and global-to-local (G2L) bidirectional interaction mechanism, this context embedding and local features continuously coevolve, enabling globally coherent multiscale feature representation.
2) We propose an angle-aware regression (AAR) strategy to address two limitations of existing regression paradigms. Numerical regression suffers from boundary discontinuity, while Gaussian-based metrics experience isotropic degeneration for square objects. By introducing explicit angular supervision into the regression loss, the AAR strategy enables robust orientation learning across arbitrary object geometries.
3) Extensive experiments on two public datasets demonstrate the superior performance of the GCAD framework. Furthermore, the effectiveness of each constituent module is validated through systematic ablation studies.

The remainder of this article is organized as follows. Section II reviews related work on oriented object

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detection (OOD) in remote sensing, multiscale feature pyramid fusion, and rotation-aware regression loss functions. Section III presents the proposed GCAD framework in detail. Section IV reports the experimental results and discussions. Section V concludes this article and outlines potential directions for future work.

II. Related Work

A. OOD in Remote Sensing

OOD in remote sensing imagery has long been a focal point of computer vision research [18], [19], [20]. Driven by the rapid advancements in deep learning, a growing number of methods have been proposed, yielding significant performance improvements. OOD is a task designed to precisely localize and identify objects with arbitrary orientations using OBBs. Unlike traditional horizontal object detection [21], [22], OOD introduces an orientation parameter, enabling the bounding boxes to tightly enclose objects with large aspect ratios or dense arrangements, thereby providing more accurate localization information. To achieve precise regression for oriented objects, existing mainstream approaches generally follow the paradigms of generic object detection, broadly divided into one-stage and two-stage pipelines.

Following the paradigm of generic one-stage detectors, such as FCOS [23] and RetinaNet [22], oriented detectors including R3Det [24] and S^2A-Net [25], along with representation-refined methods, such as Oriented RepPoints [26] and RSDet [27], directly regress OBBs from feature maps, offering high inference efficiency but remaining susceptible to feature misalignment under dense or arbitrary orientations. On the contrary, two-stage detectors, such as rotated Faster R-CNN [28], RoI Transformer [29], ReDet [30], and Oriented R-CNN [31], adopt a coarse-to-fine strategy built upon Faster R-CNN [21]. High-quality oriented proposals are first generated and then refined via rotation-invariant feature extraction. This yields superior spatial alignment at the cost of additional computation. Despite the success of CNN-based detectors, Transformer-based methods [32] have recently emerged as a promising direction, leveraging global attention mechanisms for more flexible feature aggregation. However, their high computational cost and data hunger remain open challenges in remote sensing scenarios.

B. Feature Pyramid and Multiscale Fusion

To enhance feature discrimination for multiscale objects in complex backgrounds, the FPN, pioneered by Lin et al. [6], has been widely adopted to bridge deep semantics and shallow details.

FPN variants have been proposed from diverse angles. PANet [7], FaPN [33], and LR-FPN [34] enhance bidirectional propagation and cross-scale spatial alignment. Libra R-CNN [35] and AugFPN [36] improve balanced multiscale sampling, while ScrDet [14] and HS-FPN [37] strengthen the responses of small and cluttered targets through attention and high-pass perception. BiFPN [8], ASFF [10], FE-FPN [38], and SREDet [39] introduce weighted or adaptive fusion to suppress background noise and resolve semantic conflicts. NAS-FPN [9] and DetectoRS [40] further explore topological optimization through neural architecture search and recursive feature aggregation. CISC-R [41] enables weakly supervised analysis of multiscale semantic consistency in large-scale scenes, achieving accuracy comparable to that of fully supervised methods for the first time. More recently, Qian et al. [42] integrate multiscale context aggregation with a task-consistent focal loss for OOD, leveraging cross-scale priors at the supervision level. FAAFusion [43] further introduces orientation alignment into the detector neck to improve directional consistency during cross-scale feature fusion.

However, despite these structural advancements, the FPN paradigm still suffers from intrinsic representational limitations. As illustrated in Fig. 3, traditional FPNs and their variants essentially adhere to a local-neighbor interaction pattern, propagating cross-scale information via layerwise transmission between adjacent levels and lacking a global semantic nexus that directly spans all scales. Even recent works that introduce cross-scale priors at the supervision or task-consistency level operate orthogonally to the propagation mechanism itself, leaving the long-range cross-scale information flow largely unaddressed at the feature-fusion stage. Moreover, the fusion process remains static and feed-forward, lacking dynamic refinement during hierarchical propagation. To address these limitations, our proposed framework introduces an evolving unified context embedding that drives a bidirectional cyclic interaction mechanism. Through this global semantic anchor, all pyramid levels are directly bridged, and the embedding continuously coevolves with local features during hierarchical propagation. This design eliminates both the local-neighbor bottleneck and the static-fusion limitation that constrain prior methods.

C. Angular Supervision in OOD

The evolution of regression supervision in OOD has primarily transitioned from parameterwise numerical regression to distribution-based metric learning, aiming to navigate the tradeoff between geometric precision and scale robustness.

Early detectors directly extended the horizontal regression paradigm by adding an angular parameter θ, but inevitably suffered from boundary discontinuity, as ideal predictions often jump beyond the predefined range and cause loss spikes during training [44]. To mitigate this issue, IoU-smooth L1 [14] and modular rotation loss [27] introduced loss-level constraints to soften the discontinuity. From a different perspective, Qian et al. [45] proposed to decompose the OBB regression into the joint regression of a horizontal smallest enclosing rectangle and two offsets, enabling existing HOD losses to be directly transferred to OOD. While such representation-level bridging mitigates the disparity between HOD and OOD, the resulting losses still inherit the parameter-level sensitivity of numerical regression. To fundamentally depart from regression, CSL [44] reformulated angular prediction as a classification task via circular smooth labels, effectively handling angular periodicity. Building on this paradigm, DCL [46] further improved it by densely encoding the angular errors.

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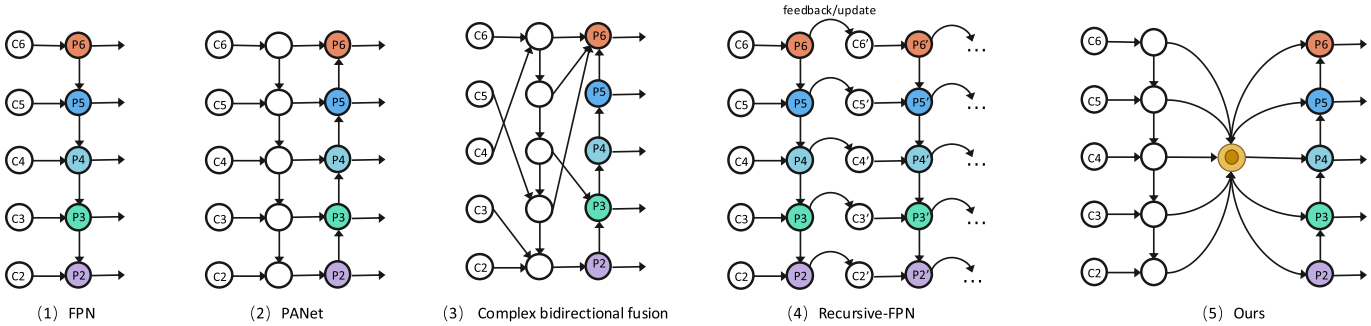


Fig. 3. Comparison of feature fusion paradigms. While traditional architectures (1)–(4) rely on relay-style local connections that cause semantic attenuation, our proposed method (5) utilizes an evolving unified context embedding to directly bridge all pyramid levels, enabling globally coherent feature representation.

In parallel with these classification-based reformulations, IoU-based optimization has emerged as another major direction, modeling bounding boxes as 2-D Gaussian distributions for boundary smoothing and scale invariance. GIoU [47] first demonstrated the advantages of directly optimizing geometric overlap metrics in general object detection, inspiring the community to explore analogous strategies for oriented variants. Since the exact IoU for rotated boxes is nondifferentiable, representative methods such as GWD [16], KLD [17], and Kalman Filter IoU (KFIoU) [48] leverage distance metrics such as Wasserstein distance and Kullback-Leibler divergence to approximate the IoU loss. By transforming box regression into a distribution distance measurement, these approaches provide an elegant mechanism for boundary smoothing and cross-scale robustness. Beyond loss design, GauCho [49] directly regresses Gaussian distributions through Cholesky decomposition to mitigate boundary discontinuity, while the angular decoding ambiguity of square-like objects remains unresolved. More recently, Hausdorff distance matching [50] introduced a geometry-aware matching strategy for rotated detection transformers, further advancing the precision of oriented regression. This trend reflects a broader shift from direct parameter regression toward metric learning, effectively bridging the gap between geometric representation and differentiable optimization.

However, this distribution-based paradigm suffers from a critical limitation, which we refer to as angular degradation. For objects with aspect ratios close to 1, such as square STOs or RAs, the corresponding Gaussian representation becomes increasingly isotropic, as shown in Fig. 2. In such cases, orientation information is substantially weakened in the covariance matrix, rendering the loss function progressively less sensitive to angular deviations. Consequently, existing methods inherently sacrifice angular supervision for specific geometries to maintain scale consistency, and no prior approach has explicitly addressed this angular degradation issue within the Gaussian-based framework. To address this, our approach explicitly integrates an angular constraint term into the metric learning objective, directly injecting orientation supervision into the Gaussian-based loss formulation. This design effectively compensates for the information loss caused by the Gaussian transformation, ensuring precise angular supervision across arbitrary geometries.

III. Proposed GCAD Method

The overall architecture of the proposed GCAD framework is illustrated in Fig. 4. Targeting the issues of restricted cross-scale semantic interaction and the loss of angular supervision induced by the isotropic degradation of square objects, the framework comprises two core components: 1) the global context-anchored cyclic (GCAC) module and 2) the AAR module. In particular, the GCAC module constructs a novel feature pyramid driven by the unified context embedding. By initializing an evolving unified context embedding and designing a bidirectional cyclic interaction mechanism, it reshapes the feature fusion paradigm, wherein the L2G pathway aggregates multiscale details to continuously update the context embedding, while the G2L pathway redistributes the global context back to local features to enhance semantic consistency. This mechanism effectively overcomes the semantic attenuation caused by relay-style transmission. Subsequently, the AAR module incorporates explicit angular supervision into the regression loss. This strategy compensates for the degradation of orientation sensitivity when the Gaussian representations of square targets become isotropic, thereby ensuring robust regression performance under extreme geometric variations.

A. GCAC FPN

To address the relay-style semantic attenuation and the lack of a unified global context identified in Section II-B, we propose the GCAC-FPN to reconstruct the multiscale feature interaction paradigm.

Unlike previous methods that treat multiscale features as independent or static nodes, we introduce an evolving unified context embedding that sequentially aggregates semantics across all pyramid levels, as depicted in the left panel of Fig. 5. At each level, this evolving embedding interacts with local features through coupled L2G and G2L cross-attention operations, mutually refining the global representation and local features. Through this bidirectional feedback, each pyramid level is directly anchored to a coherent global semantic space, ensuring consistent and robust perception of targets across drastic scale variations.

1) Bidirectional Cyclic Interaction: To overcome the relay-style information attenuation inherent in the local progressive interaction paradigm, the proposed GCAC-FPN introduces a

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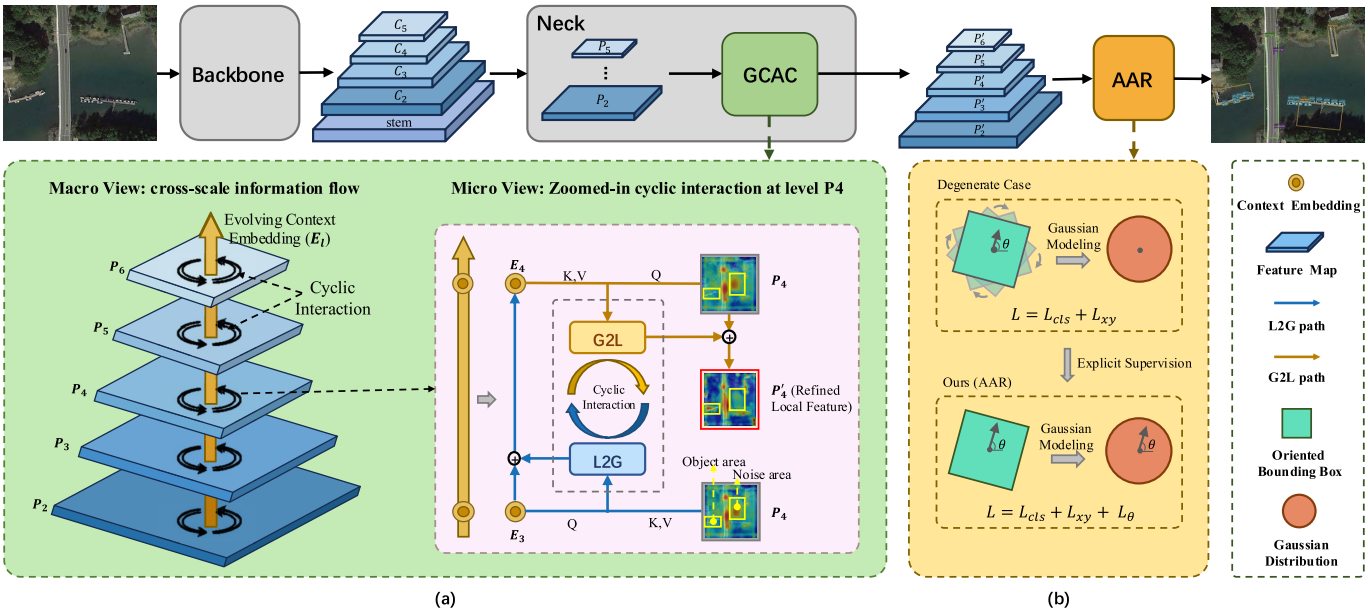


Fig. 4. Overall architecture of the proposed GCAD. Multiscale features ($\mathbf{P}_2$–$\mathbf{P}_6$) extracted by the backbone are sequentially processed by two key components. (a) The GCAC module drives a dynamic embedding-feature coevolution process to obtain refined features ($\mathbf{P}'_2$–$\mathbf{P}'_6$). Left (Macro View): A unified context embedding traverses the feature pyramid to synchronize cross-scale semantics. Middle (Micro View): At each pyramid level, the context embedding and local feature map engage in a bidirectional cyclic interaction. (b) AAR module incorporates explicit orientation supervision to predict precise arbitrary OBBs, effectively preventing the angular degradation of square objects.

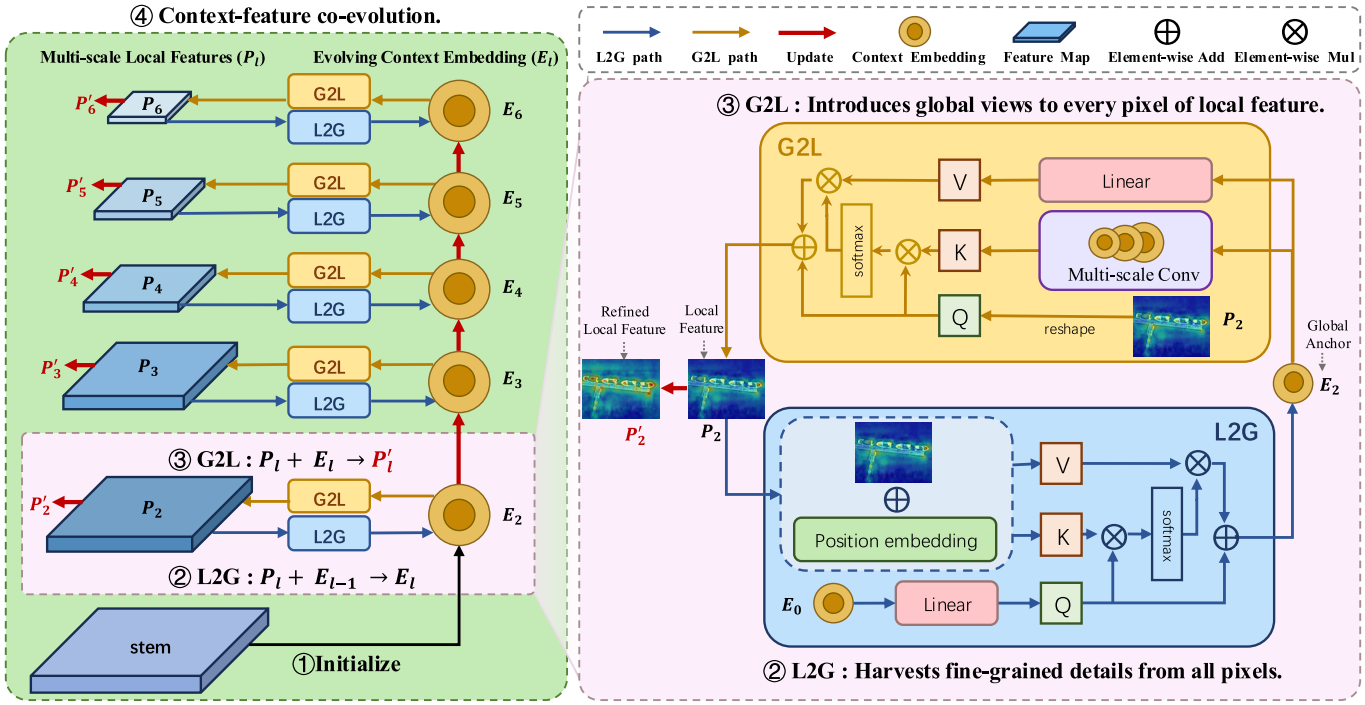


Fig. 5. Detailed interaction pipeline of the GCAC module. *Left:* The macrolevel cyclic process across all feature pyramid levels ($\mathbf{P}_2$–$\mathbf{P}_6$). Breaking away from the traditional relay-style progressive transmission, local features bypass direct spatial routing and instead evolve independently through a unified context embedding. Initialized from the stem (Step ①), this embedding sequentially aggregates cross-scale semantics to serve as an evolving global anchor. *Right:* Zoomed-in schematic of the bidirectional coevolution at a specific level l, comprising two sequential operations. In the L2G semantic aggregation (Step ②), the preceding anchor $\mathbf{E}_{l-1}$ harvests position-augmented local details to yield the updated anchor $\mathbf{E}_l$. In the G2L feature alignment (Step ③), an MSConv module extracts varying semantic granularities from $\mathbf{E}_l$ to enable scale-aware cross-attention matching, redistributing the global context back to the spatial domain to generate the aligned representation $\mathbf{P}'_l$.

traveling context embedding to drive a recurrent global alignment mechanism. By acting as a unified semantic carrier, this embedding bridges all spatial levels. This bridging, however, requires interaction in both directions. The two pathways play

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nonsubstitutable roles: L2G is the sole channel through which the context embedding accumulates cross-scale semantics, while G2L is the sole channel through which this accumulated context is delivered back to the local features. Relying on L2G alone would enrich the embedding without ever updating the local features; relying on G2L alone would broadcast a stem-initialized embedding devoid of multiscale evidence. Only their coupling closes the loop and enables true coevolution. We define a compact set of unified context embeddings, initialized from the output features of the backbone stem (corresponding to Step ① in Fig. 5), as $\mathbf{E}_0 \in \mathbb{R}^{M\times d}$, where M and d denote the sequence length and embedding dimension, respectively. The sequence length M determines the representational capacity of the unified embedding. We set $M = 64$, which encourages the embedding to concentrate on essential global context while remaining computationally efficient. A systematic analysis of this hyperparameter is provided in Section IV-B.

Rather than static fusion, the local feature maps and the context embeddings engage in a dynamic, levelwise coevolution. Let $\mathbf{F}_l \in \mathbb{R}^{H_l\times W_l\times C}$ denote the local feature map at the lth pyramid level (corresponding to $\mathbf{P}_l$ in Fig. 5), and $\mathbf{E}_{l-1}$ denote the unified context embedding propagated from the preceding level. At each stage l, the alignment process comprises two sequential operations: L2G semantic aggregation and G2L feature alignment. First, the context embedding harvests level-specific details from the local feature map (corresponding to Step ② in Fig. 5)

$$\mathbf{E}_l = \mathbf{E}_{l-1} + \mathrm{CrossAttn}_{\mathrm{L2G}}\left(\mathbf{E}_{l-1},\, \varphi\left(\mathbf{F}_l \oplus \mathbf{P}_l^{pos}\right)\right) \tag{1}$$

where $\mathrm{CrossAttn}_{\mathrm{L2G}}(\cdot)$ represents the semantic aggregation mapping, $\varphi(\cdot)$ denotes spatial flattening, $\oplus$ is elementwise addition. $\mathbf{P}_l^{\mathrm{pos}}$ denotes the spatial positional cues generated from $\mathbf{F}_l$ by a learnable 3×3 depthwise convolution. By treating the context embedding as a global query, this operation adaptively scans the position-augmented local features to distill fine-grained geometric details. This continuously evolves the unified context embedding to accumulate a globally coherent representation (detailed in Section III-A2).

Subsequently, the evolved context embedding $\mathbf{E}_l$ broadcasts this holistic prior back into the local feature space (corresponding to Step ③ in Fig. 5)

$$\mathbf{F}_l' = \mathbf{F}_l + \varphi^{-1}\left(\mathrm{CrossAttn}_{\mathrm{G2L}}\left(\varphi\left(\mathbf{F}_l\right),\, \mathrm{MSConv}\left(\mathbf{E}_l\right)\right)\right) \tag{2}$$

where $\mathrm{CrossAttn}_{\mathrm{G2L}}(\cdot)$ denotes the context injection mapping, $\varphi^{-1}(\cdot)$ is the spatial reshaping function, and $\mathrm{MSConv}(\cdot)$ is the multiscale convolution (MSConv) module. This reverse projection utilizes local features as spatial queries to retrieve relevant semantics. Crucially, MSConv dynamically decodes the context embedding to provide scale-aware key representations, ensuring that targets of varying sizes precisely match their optimal semantic granularities. Consequently, the multiscale local features are robustly anchored to the global semantic space (detailed in Section III-A3). Through this cyclic mechanism, GCAC-FPN progressively aligns features across all levels, effectively overcoming relay-style semantic attenuation.

The L2G and G2L pathways play mutually indispensable and highly complementary roles. In particular, the L2G pathway performs semantic aggregation, dynamically extracting level-specific geometric and semantic cues to drive the evolution of the context embedding; without it, the embedding would remain static and lack multiscale evidence. Conversely, the G2L pathway executes feature alignment by broadcasting the enriched global context back to anchor the local feature maps; without it, the accumulated global knowledge cannot flow back to refine the spatial domain, rendering the entire mechanism a redundant, isolated branch. Only through this bidirectional cooperation can the network form a closed loop of "aggregation-evolution-feedback" to achieve true cross-scale semantic synchronization.

2) L2G Semantic Aggregation: The L2G pathway is designed to update the unified context embedding by harvesting level-specific geometric details and semantic cues from the current spatial features. We formulate this aggregation process as a variant of cross-attention, where the context embedding acts as the query to adaptively search for salient information across the local feature map. To enable interaction between the embedding and the local feature maps, $\mathbf{F}_l$ must be flattened along the spatial dimensions. However, this spatial flattening inherently destroys the structural inductive bias. Given that precise spatial arrangement and orientation are paramount for objects in remote sensing imagery, we explicitly inject a learnable position embedding, denoted as $\mathbf{P}_l^{pos} \in \mathbb{R}^{H_l\times W_l\times C}$, into the local features before the attention operation. The position-augmented feature is then flattened into $\tilde{\mathbf{F}}_l \in \mathbb{R}^{N_l\times C}$, where $N_l = H_l \times W_l$.

In the L2G pathway, the query matrix $\mathbf{Q}$ is projected from the preceding context embedding $\mathbf{E}_{l-1}$, while the key matrix $\mathbf{K}$ and value matrix $\mathbf{V}$ are derived from the position-augmented local feature $\tilde{\mathbf{F}}_l$

$$\mathbf{Q} = \mathbf{E}_{l-1}\mathbf{W}_q \tag{3}$$

$$\mathbf{K} = \tilde{\mathbf{F}}_l\mathbf{W}_k, \quad \mathbf{V} = \tilde{\mathbf{F}}_l\mathbf{W}_v \tag{4}$$

where $\mathbf{W}_q \in \mathbb{R}^{d\times d}, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{C\times d}$ are learnable projection matrices.

The context embedding is subsequently updated by aggregating the local values through a scaled attention mechanism, followed by a residual connection to preserve the historical global context. The state update from $\mathbf{E}_{l-1}$ to $\mathbf{E}_l$ is computed as

$$\mathbf{E}_l = \mathbf{E}_{l-1} + \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}. \tag{5}$$

Intuitively, this L2G operation enables the context embedding to selectively distill critical local details from each pyramid level into a continuously updated global representation.

3) G2L Feature Alignment: Once the unified context embedding has absorbed the level-specific details via L2G, the G2L pathway broadcasts the updated global context back to the local feature map. A critical challenge in this reverse projection lies in addressing the extreme multiscale variations inherent in remote sensing imagery. Applying a uniform global representation inevitably leads to a contextual mismatch, as large objects necessitate coarse-grained high-level semantics whereas small objects rely entirely on fine-grained details. In

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a conventional cross-attention mechanism, the key matrix is derived from embeddings via standard pointwise linear projections, treating each latent embedding independently and thus failing to capture the varying semantic granularities required by multiscale local queries. To overcome this limitation, we introduce an MSConv module to dynamically decode the context embedding. Rather than relying on static pointwise projections, MSConv applies parallel 1-D convolutional layers with varying kernel sizes across the latent embedding sequence, capturing interembedding dependencies at different granularities. A smaller kernel preserves specific, fine-grained latent semantics, whereas a larger kernel aggregates a broader, coarse-grained context. These multiscale representations are subsequently concatenated and fused to form a scale-adaptive global prior.

Formally, the local feature $\mathbf{F}_l$ is flattened to generate the query matrix $\mathbf{Q}_L$, while the key matrix $\mathbf{K}_G$ and value matrix $\mathbf{V}_G$ are derived from the context embedding $\mathbf{E}_l$

$$\mathbf{Q}_L = \phi(\mathbf{F}_l)\mathbf{W}'_q \tag{6}$$

$$\mathbf{K}_G = \text{MSConv}\left(\text{Reshape}(\mathbf{E}_l)\right)\mathbf{W}'_k \tag{7}$$

$$\mathbf{V}_G = \mathbf{E}_l\mathbf{W}'_v \tag{8}$$

where $\mathbf{W}'_q \in \mathbb{R}^{C\times d}, \mathbf{W}'_k \in \mathbb{R}^{d\times d}, \mathbf{W}'_v \in \mathbb{R}^{d\times C}$ denote the learnable projection weights for the G2L pathway. Notably, the MSConv module is exclusively applied to the key matrix $\mathbf{K}_G$. This decoupled design enables scale-aware matching during the attention matching phase while preserving the original semantics within the value matrix $\mathbf{V}_G$, thereby preventing oversmoothing of the aggregated global context.

Finally, the aligned local representation $\mathbf{F}'_l$ is obtained by computing the attention weights and adding the injected global context back to the original feature map via a residual connection

$$\mathbf{F}'_l = \mathbf{F}_l + \text{Reshape}\left(\text{Softmax}\left(\frac{\mathbf{Q}_L\mathbf{K}_G^\top}{\sqrt{d}}\right)\mathbf{V}_G\right). \tag{9}$$

Through this guided alignment, the local representation $\mathbf{F}'_l$ is enriched beyond the limitations of its level-specific receptive field, effectively guided by a globally coherent semantic context that spans all pyramid levels.

B. AAR Strategy

Beyond multiscale feature representation, high-precision OOD also critically depends on a robust regression mechanism. Following the line of Gaussian-based metrics discussed in Section II-C, we adopt the state-of-the-art KFIoU as our representative baseline, which models the OBB parameterized by (x, y, w, h, θ) as a 2-D Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for boundary-smoothed similarity measurement. The baseline regression loss is formulated as $\mathcal{L}_{base} = \mathcal{L}_{xy} + \mathcal{L}_{kf}$, where $\mathcal{L}_{xy}$ optimizes the center point offset, and $\mathcal{L}_{kf} = -\ln(\text{KFIoU})$ acts as the distribution-based metric.

However, this entire family of distribution-based paradigms suffers from a critical theoretical flaw: angular degradation. To understand this mathematically, consider the covariance matrix $\boldsymbol{\Sigma}$ defined by the scaling matrix $\mathbf{S}$ and the rotation matrix $\mathbf{R}$

$$\boldsymbol{\Sigma} = \mathbf{R}\mathbf{S}\mathbf{R}^\top = \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\frac{w^2}{4} & 0\\ 0 & \frac{h^2}{4}\end{pmatrix}\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}. \tag{10}$$

Expanding this matrix multiplication yields

$$\boldsymbol{\Sigma} = \frac{1}{4}\begin{pmatrix}w^2\cos^2\theta + h^2\sin^2\theta & \left(w^2 - h^2\right)\cos\theta\sin\theta\\ \left(w^2 - h^2\right)\cos\theta\sin\theta & w^2\sin^2\theta + h^2\cos^2\theta\end{pmatrix}. \tag{11}$$

The inherent limitation of this representation becomes particularly pronounced when handling square-like objects. Mathematically, the covariance matrix degenerates into an isotropic distribution when the aspect ratio approaches one (i.e., $w \to h$)

$$\lim_{w\to h}\boldsymbol{\Sigma} = \begin{pmatrix}\frac{h^2}{4} & 0\\ 0 & \frac{h^2}{4}\end{pmatrix}. \tag{12}$$

In this state, the orientation parameter θ is mathematically canceled out, causing the angular derivative to become exactly zero ($\partial\boldsymbol{\Sigma}/\partial\theta = \mathbf{0}$). While predicting an exact $w = h$ is rare in practice, the critical issue lies in its asymptotic behavior: as the aspect ratio approaches unity, the angular gradient becomes vanishingly small. During backpropagation, this extremely weak supervision signal is easily drowned out, rendering the distribution-based metric insensitive to angular deviations for square-like targets.

To fundamentally resolve this geometric ambiguity, we design the AAR strategy. The proposed AAR strategy is illustrated in Fig. 6. Instead of discarding the elegant scale-adaptive properties of Gaussian metrics, the AAR strategy introduces an explicit angle compensation mechanism. To preserve the accurate angular gradients already provided by KFIoU for normal high-aspect-ratio objects, we introduce a hard-gating indicator function $\mathcal{G}_{sq}$. This function activates the angular compensation only when the target's aspect ratio falls within a predefined threshold $[1/\tau, \tau]$

$$\mathcal{G}_{sq} = \begin{cases}1, & \text{if } \frac{1}{\tau} \le \frac{w}{h} \le \tau\\ 0, & \text{otherwise.}\end{cases} \tag{13}$$

Empirically, the threshold τ is set to 1.05, ensuring that the angle penalty is applied exclusively to the geometric profiles susceptible to degradation. Furthermore, directly regressing the numerical angle difference often leads to loss spikes at the boundary due to the periodic nature of rotation. To guarantee a smooth optimization landscape, we encode the angle difference using a cosine-based metric that inherently accommodates the 180° (π) symmetry of OBBs. The normalized angular loss is defined as

$$\mathcal{L}_\theta = \frac{1 - \cos\left(2\left(\theta_p - \theta_t\right)\right)}{2} \tag{14}$$

where θ_p and θ_t denote the predicted and ground-truth angles, respectively.

Finally, a significant challenge in combining geometric penalties with distribution metrics is scale imbalance. A static weight for the angle loss may either overwhelm the network during early training or vanish in the later stages. To achieve adaptive gradient synchronization, we design a dynamic scale

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modulator that leverages the detached baseline loss. The final explicit angle loss $\mathcal{L}_{ang}$ is formulated as

$$\mathcal{L}_{ang} = \lambda_\theta \cdot \mathcal{G}_{sq} \cdot \mathcal{S}(\mathcal{L}_{base}) \cdot \mathcal{L}_\theta \tag{15}$$

where λ_θ is a scaling hyperparameter, and $\mathcal{S}(\cdot)$ represents the modulator defined as $\min(\text{sg}(\mathcal{L}_{base}), \gamma)$. Here, $\text{sg}(\cdot)$ denotes the stop-gradient operation, and γ acts as a clipping upper bound (set to 2.0) to prevent early-stage loss explosion. By coupling the intensity of the angle penalty to the current state of the bounding box regression, the AAR strategy forces the network to dynamically increase angular focus only when the overall localization is poor. The overall optimization objective of the regression branch is thus seamlessly unified as $\mathcal{L} = \mathcal{L}_{base} + \mathcal{L}_{ang}$.

IV. Experiments and Discussion

In this section, we evaluate the performance of the proposed GCAD. We first introduce the experimental setup, including datasets, evaluation metrics, and implementation details. Subsequently, we present experiments on two benchmark datasets to validate the performance of our method and compare it with the baselines. Furthermore, ablation studies are conducted to verify the effectiveness of our approach and discuss the specific contributions of each component. Finally, we provide a visualization analysis.

A. Experimental Setup

1) Datasets: To validate the effectiveness of the proposed method, we conducted extensive experiments on two widely used benchmarks for OBB detection in aerial images: the DIOR-R [51] dataset and the DOTA-v1.0 [52] dataset.

DIOR-R contains 23 463 optical remote sensing images and 192 518 instances, with a fixed image resolution of 800 × 800 pixels. The dataset is divided into a training-validation set (11 725 images) and a testing set (11 738 images). It covers 20 common object categories: Airplane (APL), airport (APO), BF, Babsketball court (BC), BR, chimney (CH), dam (DAM), expressway-toll-station (ETS), expressway-service-area (ESA), golf field (GF), ground track field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA), STO, tennis court (TC), train station (TS), vehicle (VE), and wind mill (WM).

DOTA-v1.0 comprises 2806 images and annotates 15 distinct object categories. These categories are abbreviated as follows: PL, baseball diamond (BD), BR, GTF, small vehicle (SV), large vehicle (LV), SH, TC, BC, STO, storage tank (ST), soccer-ball field (SBF), RA, HA, swimming pool (SP), and helicopter (HC). Due to the large scale and varying resolutions of images in the DOTA dataset, they cannot be directly fed into the network for training. Therefore, preprocessing is essential. Following the standard practice adopted in most previous works, we cropped the original images into patches of size 1024 × 1024 with an overlap of 100 pixels.

2) Evaluation Metrics: To comprehensively evaluate the performance of our method, we adopt the standard mean average precision (mAP) as the primary evaluation metric, which is widely used in object detection tasks. In particular, a predicted bounding box is considered a true positive (TP) if its IoU with the ground truth bounding box exceeds the threshold of 0.5. The overall mAP is then calculated by averaging the average precision (AP) across all object categories based on the precision-recall curve.

3) Implementation Details: To ensure a fair comparison with state-of-the-art methods, we configure the training hyperparameters to be identical or similar to those of the compared methods. All experiments are implemented using the open-source MMRotate framework [53]. In particular, we utilize the AdamW optimizer with initial learning rates of 1×10^{-4} and 1×10^{-5} for DIOR-R and DOTA-v1.0, respectively, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 0.05. The models are trained for 36 epochs with a batch size of 6. A step learning-rate schedule is adopted, with a linear warm-up during the first 500 iterations and learning-rate decay at epochs 13, 24, and 33. StripNet-S [54] is employed as the backbone and initialized with weights pretrained on the ImageNet dataset. During evaluation, the IoU threshold, confidence threshold, and nonmaximum suppression (NMS) threshold are set to 0.5, 0.05, and 0.5, respectively.

All experiments are conducted on a single NVIDIA GeForce RTX 4090 GPU with 24 GB of memory.

B. Ablation Study

In this section, we conduct a series of experiments on the DIOR-R dataset to evaluate the contribution of each proposed module and validate our architectural designs.

1) Effectiveness of Key Components: To clearly demonstrate the impact of each module, we progressively integrate the GCAC module and the AAR strategy into the baseline detector. As shown in Table I, we report the performance across comprehensive metrics, including standard AP metrics (mAP_{50}, $\text{mAP}_{50:95}$) and scale-specific metrics (mAP_s, mAP_m, mAP_l).

For a fair comparison, we use our reproduced Strip R-CNN with a standard FPN and without the proposed modules as the baseline, which achieves an mAP_{50} of 68.78%. It also serves as the efficiency reference, with 45.08-M parameters and 218.62 G FLOPs.

Effect of the GCAC Module: To validate our specific design choices, we incrementally evaluate the internal components of the GCAC module. First, introducing a basic attention mechanism, which relies on standard cross-attention without position encoding and uses naive linear projections, establishes foundational context-anchored interactions. This initial component improves mAP_{50} to 71.04%. When PE and MSConv are introduced separately, the mAP_{50} increases from 71.04% to 71.12% and 71.16%, respectively, while the corresponding $\text{mAP}_{50:95}$ improves from 46.00% to 46.27% and 46.45%, demonstrating that both components individually contribute to detection performance. Subsequently, by integrating position-aware semantic aggregation L2G and MSConv-based scale-adaptive decoding G2L, our full *GCAC* module achieves superior and more balanced representations. This synergistic design further boosts mAP_{50} to 71.26% and significantly enhances strict localization metrics, achieving an $\text{mAP}_{50:95}$ of 46.72%. Furthermore, the full GCAC yields exceptional

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TABLE I
ABLATION STUDIES OF THE PROPOSED MODULES ON THE DIOR-R DATASET

Methods	GCAC Components	Loss	mAP_{50}	$mAP_{50:95}$	mAP_s	mAP_m	mAP_l	Params (M)	FLOPs (G)
Baseline	—	Smooth L1	68.78	44.80	29.82	56.10	79.17	**45.08**	**218.62**
+ GCAC	L2G & G2L (basic)	Smooth L1	71.04	46.00	33.62	59.02	79.76	45.719	248.35
	L2G & G2L + PE	Smooth L1	71.12	46.27	33.69	58.91	79.91	45.721	248.80
	L2G & G2L + MSConv	Smooth L1	71.16	46.45	33.65	58.87	80.04	45.719	248.37
	L2G & G2L + PE & MSConv	Smooth L1	71.26	46.72	33.71	58.78	80.16	45.721	248.82
+ AAR	—	AAR	69.32	45.11	33.51	57.71	78.41	45.08	218.62
Ours (GCAD)	L2G & G2L + PE & MSConv	AAR	**71.84**	**47.03**	**34.94**	**59.24**	**80.23**	45.72	248.82

Note: Rows with "—" under GCAC Components employ a standard FPN as the neck. "L2G & G2L (basic)" denotes the bidirectional cross-attention skeleton without position embedding or multi-scale convolution. Params (M) denotes the number of learnable parameters in the model, measured in millions (10^6). FLOPs (G) represents Floating-Point Operations, which reflects the computational complexity of the model, measured in billions (10^9).

TABLE II
CROSS-DATASET ABLATION STUDIES ON THE DOTA-V1.0 DATASET

Methods	mAP_{50}	mAP_{75}	$mAP_{50:95}$
Baseline	80.06	54.30	49.79
+ GCAC	80.59	55.04	50.57
+ AAR	80.25	54.90	50.39
Ours (GCAD)	**80.72**	**55.27**	**50.89**

large object detection (mAP_l: 80.16%), confirming that our tailored architectural improvements effectively align global context with local features across diverse scales. The entire GCAC formulation introduces only a moderate computational overhead compared to the baseline (+0.64-M Params, +30.2-G FLOPs), demonstrating a highly favorable accuracy-efficiency tradeoff.

Effect of the AAR Strategy: Integrating the AAR strategy alone into the baseline improves mAP_{50} from 68.78% to 69.32%, while significantly boosting small object detection (mAP_s: 29.82% → 33.51%) and medium object detection (mAP_m: 56.10% → 57.71%). Crucially, this robust improvement is achieved with zero additional parameters and FLOPs during inference, as the AAR acts purely as a loss function modifier to provide explicit angular supervision. This validates that our cosine-based angle loss with soft modulation effectively compensates for orientation degradation in square-like targets without increasing the inference burden.

Full Model GCAD: Combining both modules yields the complete GCAD framework, which achieves the best overall performance with 71.84% mAP_{50} and 47.03% $mAP_{50:95}$. The improvement in small object detection is particularly noteworthy, increasing by 5.12% over the baseline (mAP_s: 29.82% → 34.94%). This demonstrates that the two modules are highly complementary: GCAC-FPN provides globally coherent multiscale features that preserve spatial details, while the AAR strategy ensures robust angular supervision for geometrically ambiguous targets. Ultimately, GCAD maintains a reasonable computational footprint (45.72-M parameters, 248.82-G FLOPs), confirming that GCAD achieves an optimal balance between detection accuracy and model efficiency.

Cross-Dataset Validation on DOTA: To further verify the cross-dataset effectiveness of the proposed modules, we conduct a structured ablation on the DOTA-v1.0 dataset (Table II). Adding GCAC alone yields gains of + 0.53% mAP_{50}, + 0.74% mAP_{75}, and + 0.78% $mAP_{50:95}$ over the baseline, confirming the cross-dataset utility of the GCAC interaction. The smaller mAP_{50} gain of GCAC on DOTA (+0.53%) compared with that on DIOR-R (+2.48%) may be partly attributed to the relatively strong DOTA baseline (80.06% mAP_{50}), which leaves less room for further improvement at the IoU = 0.5 threshold. Differences in dataset characteristics may also contribute to the variation in performance gains. Meanwhile, the larger gains in mAP75 (+0.74%) and $mAP_{50:95}$ (+0.78%) indicate that the benefit of GCAC becomes more evident under stricter IoU criteria. AAR contributes additional gains of +0.19%, +0.60%, and +0.60%, with the improvements becoming more pronounced under stricter IoU thresholds, which is consistent with AAR's role in angular precision refinement. The full GCAD framework achieves the largest gains of +0.66%, +0.97%, and +1.10% on the three metrics.

2) Feature Activation Visualization of GCAC: To visually validate the effectiveness of the GCAC module in aligning multiscale features, we extract and compare the spatial feature map activations across pyramid levels P_2 through P_6, as illustrated in Fig. 7. Instead of employing gradient-based approximation methods, we extract the spatial activation maps directly and apply progressive Gaussian smoothing to the higher-level features to ensure a clear visual presentation.

First, we examine the low-level features at P_2 and P_3, which are primarily responsible for capturing fine-grained texture details and geometric boundaries of small targets. The selected scene contains densely moored small SHs and narrow pier edges. While the baseline heatmaps successfully capture the general locations of these small instances, their activation responses are relatively weak and exhibit a certain degree of diffusion across adjacent objects. On the contrary, our method significantly intensifies the feature responses and produces much sharper spatial activations. As specifically highlighted in the zoomed-in details of Area A and Area B, the high-response regions of our proposed model more precisely delineate the boundaries of the densely clustered targets. This indicates that the GCAC module, driven by its L2G detailed

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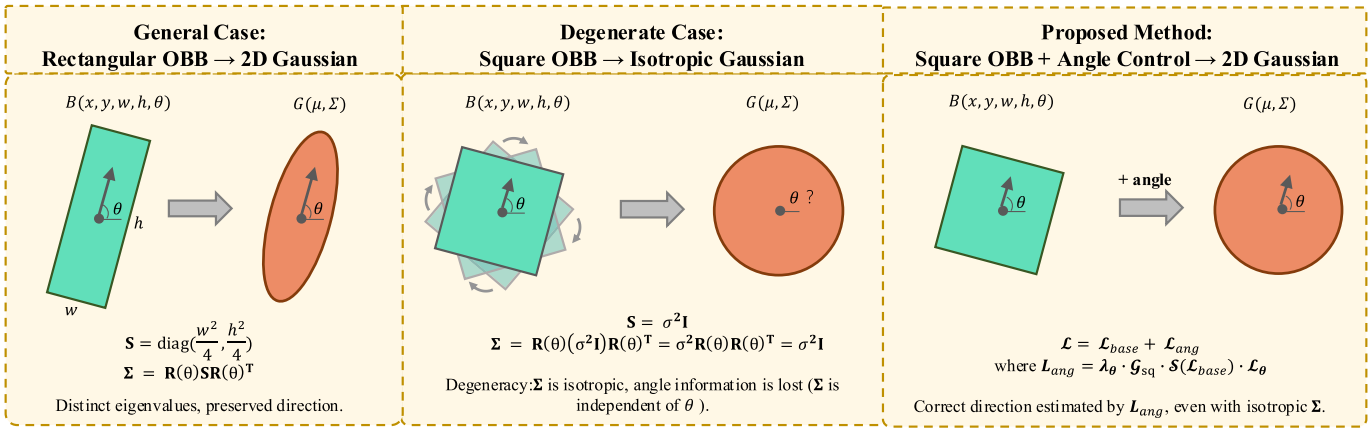


Fig. 6. Addressing angular degeneracy in Gaussian-based representations. To overcome the loss of orientation cues when square targets collapse into isotropic Gaussians, our AAR introduces an explicit angular penalty ($\mathcal{L}_{ang}$) to ensure robust direction estimation.

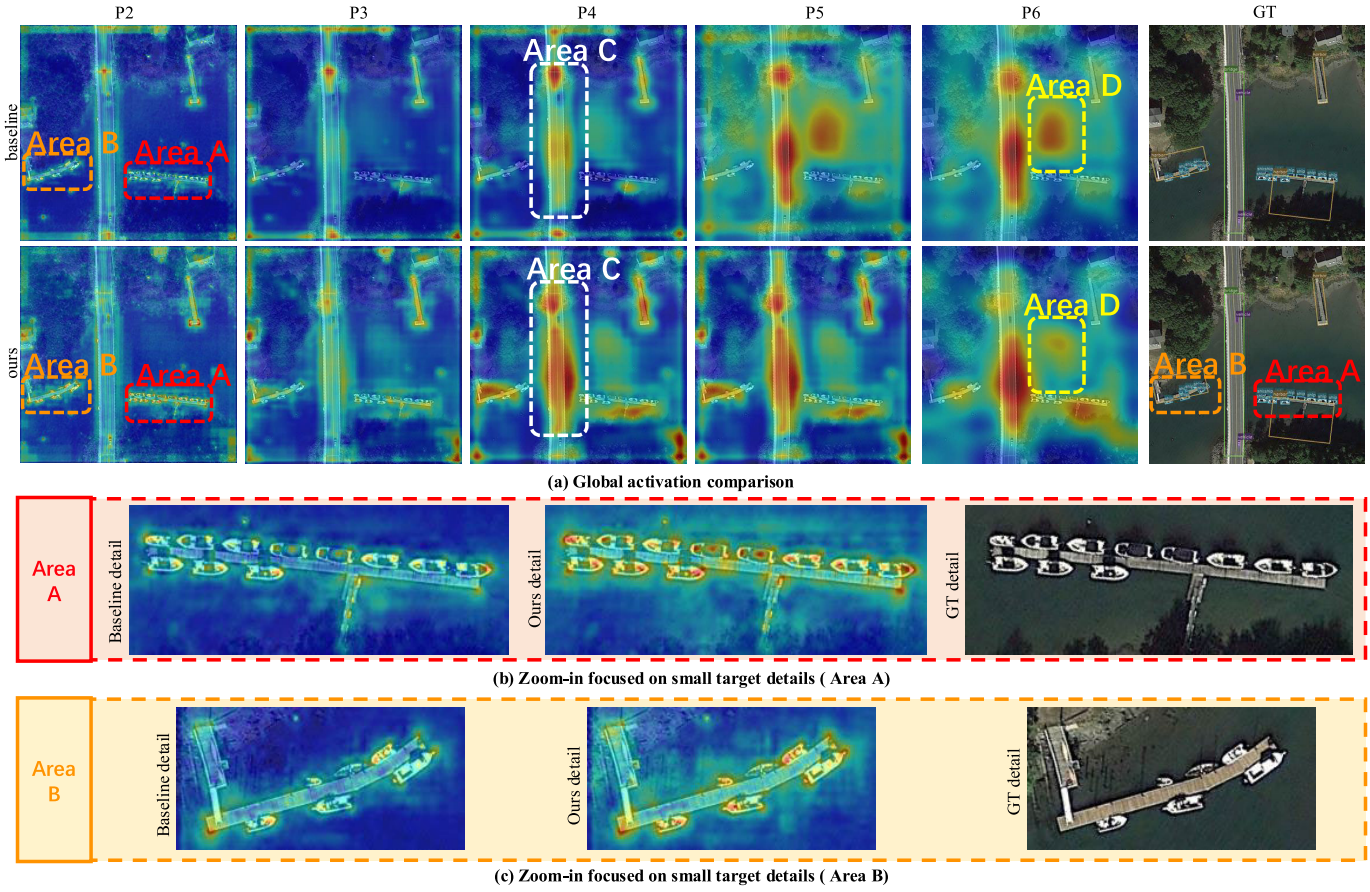


Fig. 7. Visualization of feature heatmaps across pyramid levels ($P_2 - P_6$). (a) Global activation comparison. Compared to the baseline, our method maintains coherent semantic responses for large objects across multiple scales (Area C) while effectively suppressing background noise in deep layers (Area D). (b) and (c) Zoomed-in views of Area A and Area B. Driven by the GCAC module, our method significantly enhances fine-grained representations, yielding sharper and more distinct activations for small, densely clustered targets.

harvesting, effectively mitigates the information loss caused by downsampling and preserves rich spatial details for small objects.

Moving to the mid-level features at P_4 and P_5, these layers are expected to capture the holistic structure of larger objects and their immediate contextual information. As observed in Area C highlighting a long BR, the baseline model exhibits a strong response only at a local extremity (the top segment), leaving the extended main body with severely weak and fragmented activations. Conversely, our method significantly expands the high-response coverage along the main body of the BR. Our model successfully preserves the overall structural integrity to a much greater degree than the highly fragmented baseline. This demonstrates that the dynamic context injected

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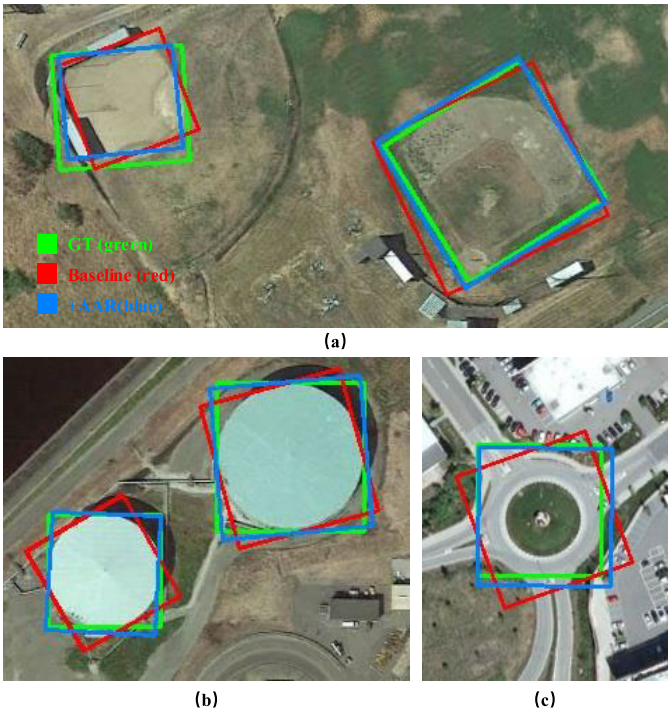


Fig. 8. Visual ablation of the AAR module on three geometrically symmetric categories: (a) BF, (b) STO, and (c) RA. Green: ground-truth; red: baseline; blue: baseline + AAR.

by the GCAC module enables the network to perceive a more complete morphology of large targets, mitigating the severe bias toward isolated local parts.

Finally, we observe the highest-level feature at P_6, which contains the strongest semantic information and is crucial for global localization. The activation of the baseline in this layer is highly ambiguous, suffering from severe spurious activations in the background regions. As specifically highlighted in Area D, the baseline produces extensive and diffuse high-response regions over the open water. Conversely, our method ensures that the semantic focus remains precisely anchored to the valid targets. The background interference in Area D is substantially attenuated, effectively reducing the false responses in nontarget regions.

This visual evidence strongly corroborates our core motivation, proving that the GCAC module successfully overcomes the restricted cross-scale interaction inherent in traditional relay-style paradigms. By facilitating a more comprehensive and synchronized bidirectional information flow across all levels, it ensures that small targets retain sharp geometric boundaries in shallow layers, while in deep layers, the global semantic focus is highly concentrated on valid targets, effectively restraining background interference.

3) Visualization of AAR: Fig. 8 visualizes the bounding box predictions of the baseline and the baseline equipped only with the AAR module (Baseline + AAR) on three representative categories whose aspect ratios are close to unity: BF, STO, and RA. On all three categories, the baseline (red) shows a visible angular offset from the ground-truth boxes (green), whereas the Baseline + AAR predictions (blue) follow the ground-truth orientations with smaller deviations. This contrast is consistent with the angular degradation issue discussed in Section II-C, where the isotropic Gaussian representation of square-like targets eliminates the orientation gradient in the covariance matrix.

4) Effectiveness of the Proposed Loss Function: To further evaluate the effectiveness of our proposed loss function, we conducted comparative experiments on the DIOR-R dataset using various bounding box regression losses. All comparative experiments reported in Table III are built upon the StripNet-S [54] + GCAC-FPN architecture for a fair comparison.

As shown in Table III, the baseline model utilizing the standard Smooth L1 [13] loss achieves an mAP_{50} of 71.26%. We compare our method with several advanced loss functions designed for OOD, including GWD [16], KLD [17], and KFIoU [48]. While these distribution-based loss functions yield marginal improvements in mAP_{50} over the Smooth L1 baseline (e.g., GWD improves mAP_{50} to 71.37%), our proposed loss demonstrates strong performance across multiple evaluation metrics. In particular, our method achieves an mAP_{50} of 71.84%, outperforming Smooth L1 and the second-best method, GWD, by 0.58 and 0.47 percentage points, respectively. This competitive performance across multiple evaluation criteria validates the effectiveness of our approach. Notably, our method achieves the best mAP_{75}, indicating improved localization accuracy under a stricter IoU threshold. This result further supports the effectiveness of the tailored angle compensation in improving orientation estimation for geometrically challenging targets.

To further justify the design choices within the AAR loss, we conducted ablations on the formulation of the angle term $\mathcal{L}_\theta$ in (14) and the dynamic modulator $\mathcal{S}(\cdot)$ in (15). Results are summarized in Table IV, with each variant replacing only the corresponding component while keeping all other settings fixed. The two alternative angle term formulations both fail to train stably due to boundary discontinuity and periodicity mismatch, respectively. The modulator variants yield measurable performance drops from the adopted formulation. Together, these results confirm the necessity of each design choice in the AAR loss.

5) Impact of the Aspect-Ratio Threshold τ in AAR: The threshold τ in (13) controls which targets are routed through the angular compensation branch of AAR. We evaluate four values, $\tau \in \{1.02, 1.05, 1.10, 1.15\}$, while keeping all other settings fixed and using Baseline + AAR (without GCAC) as the test configuration. As shown in Table V, the detection accuracy exhibits an inverted-U dependency on τ and attains the highest 69.32% mAP at $\tau = 1.05$. The accuracy difference across the evaluated range remains within 0.64% mAP, indicating that $\tau = 1.05$ represents the empirical optimum on a relatively stable operating range.

6) Impact of the Sequence Length of Context Embedding: As illustrated in Fig. 9, we conduct an ablation study on the DIOR-R dataset to thoroughly investigate the impact of the context embedding length M on detection performance and computational efficiency. The accuracy curve exhibits a distinct inverted U-shape, peaking precisely at $M = 64$. When M is relatively small, the restricted sequence length creates a representational bottleneck. It is insufficient to fully

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TABLE III
COMPARISON OF DIFFERENT LOSS FUNCTIONS ON THE DIOR-R DATASET

Backbone + Neck	Loss Function	mAP_{50}	mAP_{75}	$mAP_{50:95}$	mAP_s	mAP_m	mAP_l
StripNet + GCAC-FPN	SmoothL1 [13]	71.26	47.41	46.72	33.71	58.78	80.16
	GWD [16]	71.37	48.80	**47.80**	34.42	59.48	78.72
	KLD [17]	71.28	48.87	45.69	34.42	**59.76**	80.22
	KFIoU [48]	71.33	48.42	46.87	34.24	58.14	78.80
	Ours(AAR)	**71.84**	**49.04**	47.03	**34.94**	59.24	**80.23**

TABLE IV
ABLATION STUDY OF THE ANGLE TERM FORMULATION $\mathcal{L}_\theta$ (14) AND THE DYNAMIC MODULATOR $\mathcal{S}(\cdot)$ (15) ON DIOR-R

Component	Variant	mAP_{50}	$mAP_{50:95}$
$\mathcal{L}_\theta$	$\lvert\theta_p - \theta_t\rvert/(\pi/2)$	26.28†	–
	$(1-\cos(\theta_p-\theta_t))/2$	51.50†	–
$\mathcal{S}(\cdot)$	1 (constant)	68.25	43.24
	$\sigma(\mathcal{L}_{base})$	69.02	44.74
	Ours	**69.32**	**45.11**

† Variants diverge during training; the reported value is the best mAP_{50} obtained before divergence.

TABLE V
IMPACT OF THE ASPECT-RATIO THRESHOLD τ ON DIOR-R

τ	mAP_{50} (%)	ΔmAP
1.02	69.14	−0.18
1.05	**69.32**	—
1.10	68.99	−0.33
1.15	68.68	−0.64

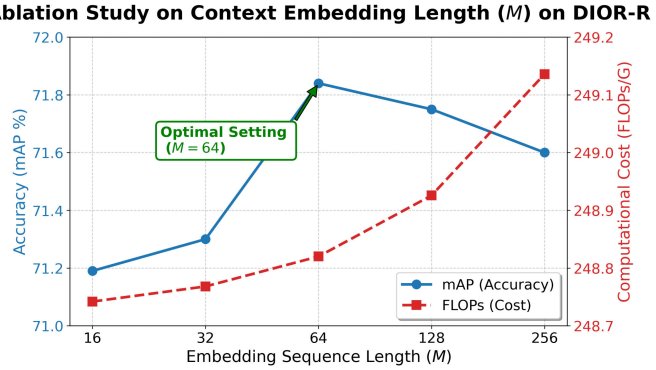


Fig. 9. Impact of the context embedding length M on detection accuracy and computational cost evaluated on the DIOR-R dataset.

encode the rich semantic diversity and complex spatial layouts inherent in large-scale remote sensing imagery. Consequently, the impoverished global representation fails to provide adequate semantic guidance during the G2L interaction, yielding suboptimal detection accuracy. As M increases to 64, the context embedding achieves an adequate capacity to capture comprehensive full-scale context without incorporating excessive noise, resulting in the highest mAP of 71.84%. At this optimal point, the global and local features are well-aligned through the cyclic interactions. However, further expanding the sequence length to 128 or 256 leads to a gradual performance degradation. We attribute this decline to two primary factors: First, an overly extended sequence introduces information redundancy and inadvertently encodes background noise into the global representation. Second, within the cross-attention mechanism, an excessively long embedding sequence exacerbates attention dilution, making it difficult for the network to focus on the most discriminative foreground features during optimization. Furthermore, from an efficiency perspective, the red dashed line indicates that the computational cost increases only marginally across the entire spectrum of M. This negligible overhead validates the efficiency of our context-anchored interaction design. Consequently, setting $M = 64$ strikes an ideal balance between high-precision detection and computational economy, and it is thus adopted as the default configuration for the GCAD framework.

Beyond the inverted U-shape observed on DIOR-R, a question naturally arises: is the optimal M coupled with the input image resolution, particularly when transferring to DOTA (1024×1024) from DIOR-R (800×800)? We argue that the choice of M is primarily governed by the spatial scale of the feature pyramid rather than the raw input resolution. After backbone down-sampling, the deepest feature maps for the two datasets differ only modestly (on the order of 25×25 on DIOR-R and 32×32 on DOTA), so a context length of $M = 64$ that captures sufficient global semantics on one resolution remains adequate on the other. This is empirically corroborated by the state-of-the-art results obtained on DOTA in Section IV-C using the same $M = 64$. Nevertheless, for substantially larger inputs (e.g., 2048 +), where pyramid features carry richer spatial layouts and broader global context, M may require revisiting; investigating a resolution-aware or dynamically adjusted M is therefore left as a direction for future work.

C. Comparison Experiments

In this section, we compare the performance of the proposed GCAD with several representative methods on the DIOR-R and DOTA datasets.

1) Comparison Experiments on the DIOR-R Dataset: Experimental results demonstrate that the proposed GCAD outperforms state-of-the-art methods, further validating the effectiveness of our approach. Table VI presents the quantitative comparison results on the challenging DIOR-R dataset. As observed, the proposed GCAD method achieves a superior mAP of 71.84%, outperforming all other compared methods listed in the table. In particular, GCAD surpasses the second-best method, RVSA (70.67%), by 1.17 percentage points, and outperforms SARFA-Net (68.90%) and CPMFNet

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TABLE VI
PERFORMANCE COMPARISONS ON THE DIOR-R DATASET

Method	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP(%)
R-RetinaNet [22]	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	77.71	67.57	61.09	81.46	47.33	38.01	60.24	57.55
CSL [44]	68.02	25.33	76.12	81.40	21.33	72.71	19.13	73.30	54.74	74.32	77.92	29.42	37.62	79.44	68.52	64.32	81.44	44.83	41.34	63.62	57.74
FR OBB [21]	62.79	26.80	71.72	80.91	34.20	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	71.21	81.44	47.31	50.46	65.21	59.54
SASM [55]	61.41	46.03	73.22	82.04	29.41	71.03	30.63	53.91	69.22	70.04	77.02	39.33	47.51	78.62	66.14	62.92	79.93	54.41	40.62	63.01	59.82
Gliding Vertex [56]	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
ROI Trans [29]	63.12	25.34	71.83	81.33	36.04	72.62	20.33	75.32	68.44	68.92	82.74	37.72	52.73	81.33	73.42	70.71	81.53	45.32	51.12	66.24	61.30
R^3Det [24]	67.10	41.63	74.34	81.33	35.61	74.72	20.82	76.33	62.34	72.31	78.14	31.92	52.81	80.42	70.43	68.63	81.54	51.51	45.52	63.93	61.55
O-RCNN [31]	63.13	36.73	72.11	81.32	44.33	72.72	26.94	77.70	67.91	68.72	83.11	40.93	57.74	**89.51**	73.65	71.13	81.41	53.81	51.10	66.10	64.05
S^2A-Net [25]	67.98	44.44	71.63	81.39	42.66	72.72	27.08	70.40	79.03	75.56	81.02	43.41	56.45	81.12	68.00	70.03	87.07	53.88	51.12	65.31	64.50
DODet [57]	63.40	43.35	72.11	81.32	43.12	72.59	33.32	78.77	70.84	74.15	75.47	48.00	59.31	85.41	74.04	**71.56**	81.52	55.47	**51.86**	66.40	65.10
CPMFNet [58]	71.75	46.19	78.81	**90.03**	45.01	78.19	34.53	69.27	78.83	76.58	83.29	45.59	58.91	81.09	79.81	69.88	88.98	59.42	48.58	65.91	67.58
Strip R-CNN [54]	71.73	48.85	80.08	89.87	45.86	78.29	38.87	85.91	73.42	78.10	83.39	46.24	59.38	81.18	74.48	70.50	89.70	65.70	49.00	65.07	68.78
SARFA-Net [59]	71.80	**53.60**	77.80	89.50	43.90	78.60	37.30	71.20	87.40	79.00	83.30	49.50	58.30	88.40	72.00	70.10	89.70	61.90	49.70	65.70	68.90
RVSA [60]	**81.10**	48.86	**81.13**	88.27	**51.02**	79.91	39.12	74.69	**88.60**	77.83	83.53	46.48	**64.14**	81.19	**84.05**	71.47	**89.97**	65.73	50.58	65.62	70.67
Ours (GCAD)	80.70	52.50	80.10	89.90	50.20	**80.90**	**43.50**	**88.20**	77.00	**79.10**	**85.30**	**51.60**	63.40	81.20	83.70	70.20	89.60	**65.90**	50.20	**73.70**	**71.84**

TABLE VII
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.0 DATASET

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP(%)
Single-scale methods																
EMO2-DETR [61]	87.99	79.46	45.74	66.64	78.90	73.90	73.30	90.40	80.55	85.89	55.19	63.62	51.83	70.15	60.04	70.91
CenterMap [62]	89.02	80.56	49.41	61.98	77.99	74.19	83.74	89.44	78.01	83.52	47.64	65.93	63.68	67.07	61.59	71.59
AO2-DETR [63]	86.01	75.92	46.02	66.65	79.70	79.93	89.17	90.44	81.19	76.00	56.91	62.45	64.22	65.80	58.96	72.15
DRN [64]	89.71	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	69.30	69.63	58.48	73.23
SASM [55]	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37	74.92
Gliding Vertex [56]	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
S^2A-Net [25]	89.07	82.22	53.63	69.88	**80.94**	82.12	88.72	90.73	83.77	86.92	63.78	67.86	76.51	73.03	56.60	76.38
R^3Det-KLD [17]	88.90	84.17	55.80	69.35	78.72	84.08	87.00	89.75	84.32	85.73	64.74	61.80	76.62	78.49	70.89	77.36
LSKNet-S [65]	89.66	85.52	57.72	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82	77.49
CPMFNet [58]	89.53	82.92	54.63	77.58	79.37	84.87	88.23	**90.95**	86.95	84.93	68.20	70.39	76.35	72.01	66.68	78.23
PKINet-S [66]	89.72	84.20	55.81	**77.63**	80.25	84.45	88.12	90.88	87.57	86.07	66.86	70.23	77.47	73.62	62.94	78.39
RTMDet-R [67]	89.43	84.21	55.20	75.06	80.81	84.53	88.97	90.90	87.38	87.25	63.09	67.87	78.09	80.78	69.13	78.85
Strip R-CNN-S [54]	88.91	86.38	57.44	76.37	79.73	84.38	88.25	90.86	86.71	87.45	**69.89**	66.82	**79.25**	**82.91**	**75.58**	80.06
Ours(GCAD)	**90.02**	**87.25**	**60.32**	75.54	80.45	**87.10**	**90.02**	90.81	**88.24**	**87.65**	68.22	**71.35**	78.22	80.94	74.74	**80.72**
Multi-scale methods																
CSL [44]	**90.25**	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93	76.17
DODet [57]	89.96	85.52	58.01	81.22	78.71	85.46	88.59	90.89	87.12	87.80	70.50	71.54	82.06	77.43	74.47	80.62
AOPG [51]	89.88	85.57	60.90	81.51	78.70	85.29	88.85	90.89	87.60	87.65	71.66	68.69	82.31	77.32	73.10	80.66
KFIoU [48]	89.44	84.41	62.22	82.51	80.10	86.07	88.68	90.90	87.32	**88.38**	72.80	71.95	78.96	74.95	75.27	80.93
PKINet-S [66]	89.02	**86.73**	58.95	81.20	80.41	84.94	88.10	90.88	86.60	87.28	67.10	**74.81**	78.18	81.91	70.62	81.06
RTMDet-R [67]	88.01	86.17	58.54	82.44	81.30	84.82	88.71	90.89	**88.77**	87.37	71.96	71.18	81.23	81.40	77.13	81.33
RVSA [60]	88.97	85.76	61.46	81.27	79.98	85.31	88.30	90.84	85.06	87.50	66.77	73.11	84.75	81.88	77.58	81.24
LSKNet-S [65]	89.57	86.34	63.13	83.67	82.20	86.10	88.66	90.89	88.41	87.42	71.72	69.58	78.88	81.77	76.52	81.64
Strip R-CNN-S [54]	89.17	85.57	62.40	**83.71**	81.93	86.58	88.84	90.86	87.97	87.91	72.07	71.88	79.25	82.45	**82.82**	82.28
Ours (GCAD)	90.21	86.34	**67.58**	78.92	**82.41**	**87.15**	**91.35**	**90.93**	86.42	87.95	**76.81**	71.50	**85.64**	**83.21**	77.23	**82.91**

(67.58%) by 2.94 and 4.26 percentage points, respectively. In terms of category-specific performance, GCAD achieves the leading AP in eight out of the 20 categories (i.e., CH, DAM, ETS, GF, GTF, HA, TS, and WM). Notably, our method demonstrates significant advantages in challenging categories. For instance, in the WM category, GCAD achieves an AP of 73.70%, surpassing the second-best result (DODet) by 7.30 percentage points. Similarly, for TS, our method obtains a substantial score of 65.90%. Furthermore, even in categories where GCAD does not rank first (e.g., BC and TC), it remains highly competitive, trailing the top results by only marginal differences. These results confirm the effectiveness and robustness of GCAD in OOD tasks.

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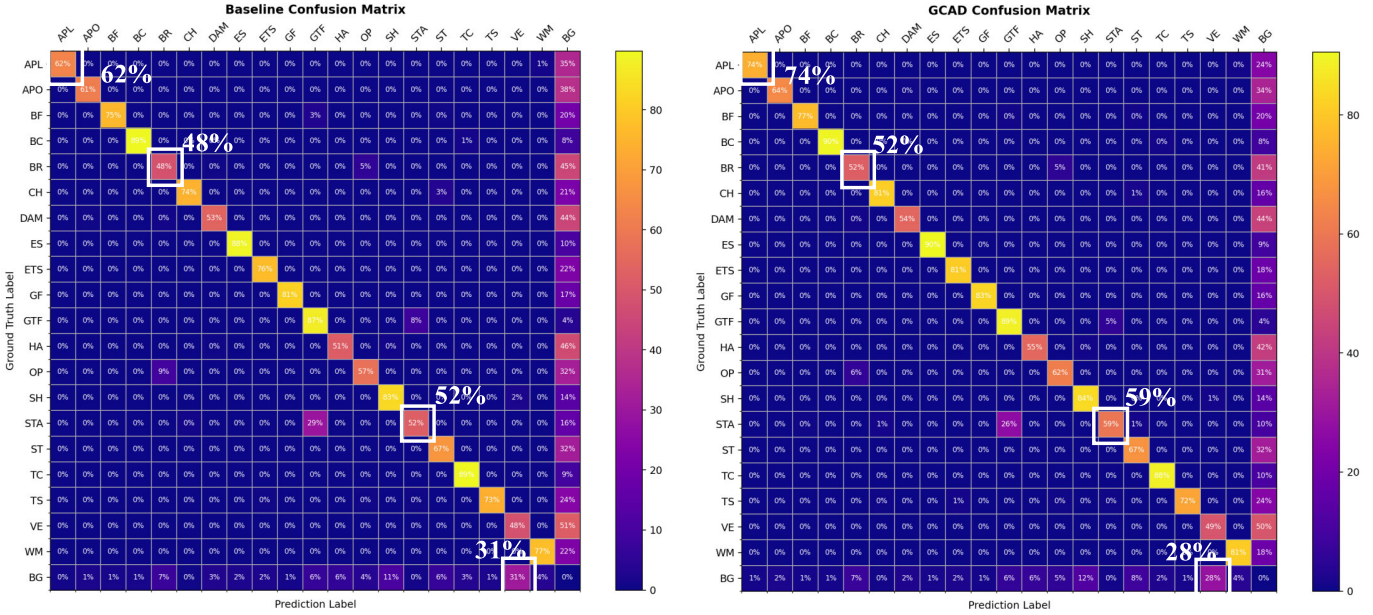


Fig. 10. Normalized confusion matrices of the baseline and GCAD on the DIOR-R dataset.

2) Comparison Experiments on the DOTA Dataset: Table VII presents the performance comparison of different methods on the DOTA dataset. Experimental results demonstrate that the proposed GCAD outperforms existing methods in terms of detection accuracy, further validating its effectiveness. In particular, under the single-scale testing protocol, GCAD achieves an mAP of 80.72%, surpassing advanced models such as Strip R-CNN-S (80.06%), RTMDet-R (78.85%), and PKINet-S (78.39%). This result demonstrates the effectiveness of GCAD in extracting discriminative features under the single-scale inference setting. Furthermore, with multiscale training and testing, the performance of GCAD is further improved to 82.91% mAP. This result outperforms the most competitive multiscale methods, including LSKNet-S (81.64%), RVSA (81.24%), and Strip R-CNN (82.28%). These comprehensive comparisons confirm that GCAD maintains leading detection capabilities even when confronted with the extremely complex scenes and diverse object categories present in the DOTA dataset.

D. Further Analysis and Visualization

1) Classwise Discriminative Analysis: Fig. 10 compares the normalized confusion matrices of the Baseline and GCAD on the DIOR-R dataset. Observing the diagonal elements, GCAD achieves significant improvements across targets of various scales and morphologies. For extremely small-scale targets like APL, the accuracy increases substantially from 62% to 74%, verifying the effective retention of fine-grained features. The accuracy for BR also improves from 48% to 52%. Notably, for STA, a typical square-like target, the accuracy rises from 52% to 59%, successfully validating our optimized design for square objects. Furthermore, GCAD enhances robustness against background interference. As highlighted, the false positive rate of misclassifying background as VE is reduced from 31% to 28%, which benefits from the aggregation of multiscale contextual information.

Two limitations remain visible in the confusion matrix. First, residual cross-classification persists between visually similar categories sharing comparable geometric layouts, such as STA and ground track field, where disambiguating fine-grained semantic cues beyond geometric differences remains challenging. Second, although background-to-VE false positives have been reduced from 31% to 28%, densely distributed SVs in cluttered backgrounds still occasionally trigger spurious activations.

2) Visualization of Detection Results: To intuitively demonstrate the effectiveness of our proposed method, we visualize the detection results on both the DIOR-R and DOTA datasets. As shown in Fig. 11, GCAD exhibits strong robustness against extreme scale variations in optical aerial imagery. Benefiting from its multiscale feature representation, the model can simultaneously localize large-scale targets (e.g., APOs, dams, and HAs) and densely packed small objects (e.g., VEs and SHs) with minimal missed detections. Building upon this robust localization, Fig. 12 further illustrates our model's capability in predicting precise oriented bounding boxes within highly cluttered environments. The zoomed-in regions specifically highlight GCAD's ability to precisely localize closely moored SHs and densely arranged STOs against complex backgrounds. Collectively, these visualizations confirm that our approach effectively handles extreme scale variations, arbitrary orientations, dense distributions, and severe background interference.

3) Inference Speed Analysis: To complement the FLOPs-based complexity analysis, we further evaluate the actual inference speed on the DIOR-R dataset. All measurements are conducted with a batch size of 1 on the inference platform specified in Section IV-A, and the results are reported in Table VIII. The comparison reveals a clear accuracy-efficiency tradeoff across methods: one-stage detectors

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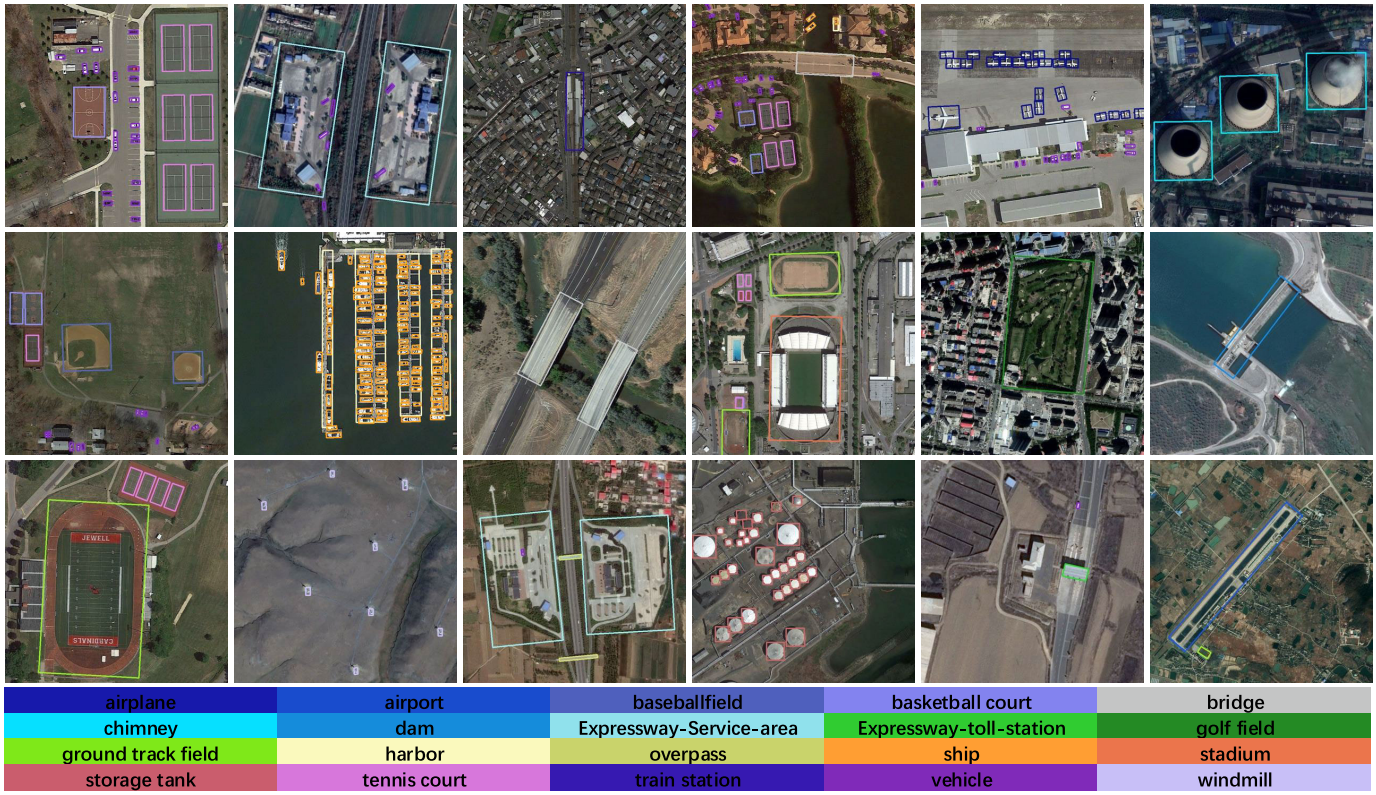


Fig. 11. Visualization results of the proposed GCAD on the DIOR-R dataset.



Fig. 12. Visualization results of the proposed GCAD on the DOTA dataset.

(R^3Det, S^2A-Net, LSKNet-S) achieve higher throughput at the cost of lower accuracy, while two-stage detectors operate in the 31–45 FPS range with substantially better localization performance. Within the same speed tier as PKINet-S + O-RCNN (34.4 FPS), GCAD reaches 31.0 FPS while delivering an impressive improvement of 4.81 mAP

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TABLE VIII
INFERENCE SPEED COMPARISON ON DIOR-R DATASET

Method	Type	FPS(img/s)	$\mathrm{mAP}_{50}(\%)$
R^3Det [24]	One-stage	54.9	61.55
S^2A-Net [25]	One-stage	**64.3**	64.50
LSKNet-S [65]	One-stage	59.1	65.90
PKINet-S + O-RCNN [66]	Two-stage	34.4	67.03
Strip R-CNN [54]	Two-stage	45.7	68.78
GCAD (Ours)	Two-stage	31.0	**71.84**

points. Compared with Strip R-CNN, GCAD achieves 71.84% mAP_{50} at 31.0 FPS, reflecting the tradeoff between detection accuracy and inference speed. The sequential dependency of the context embedding across pyramid levels, together with the L2G-to-G2L dependency within each level, limits full interlevel parallelization and may partly account for the additional inference latency.

V. CONCLUSION

In this article, we proposed GCAD, a novel method, to address the challenges of extreme scale variation and the loss of angular supervision. In particular, we introduced the GCAC-FPN, which redesigns the multiscale feature interaction paradigm. By employing a global context embedding as a centralized evolving global anchor, the proposed G2L and L2G bidirectional cyclic interaction mechanisms effectively overcome the semantic attenuation inherent in traditional relay-style feature pyramids, ensuring synchronized feature evolution across all scales. Furthermore, to tackle the localization inaccuracy of square targets, the AAR module introduces explicit angular supervision into the regression loss. This design mitigates the orientation insensitivity caused by isotropic Gaussian representations, enabling more accurate localization of square objects. Experimental results on the DOTA and DIOR-R benchmarks demonstrate that GCAD achieves state-of-the-art performance, particularly in accurately localizing objects with extreme scales and complex geometries. Our work provides a new perspective on constructing global semantic hubs and angle-sensitive supervision mechanisms for high-performance Earth observation tasks. Moving forward, we plan to streamline the GCAD architecture for lightweight deployment, enabling real-time detection on resource-constrained platforms, such as on-board processors of UAVs and edge devices for in-orbit intelligent interpretation of satellite imagery. Extension toward multimodal sensing and cross-domain generalization is also of interest for future investigation.

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